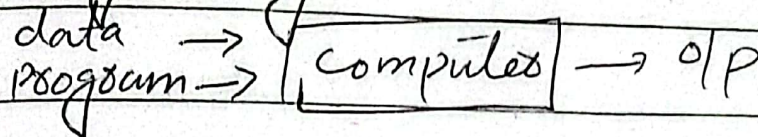
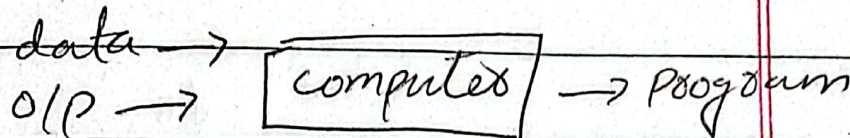


ML - field of Computer Science that uses statistical techniques to give Computer systems the ability to "learn" data without being explicitly programmed.

Traditional programming:



ML:



→ give data to algorithm

→ explore on the data

→ identify patterns b/w i/p & o/p

Why use ML if scenario is not given?

1. You cannot write code for scenario.
ex: spam or email

2. we cannot imagine no of cases.

ex: image classification

3. Uses in data mining / where data can not be search using graphs

data analysis → extract patterns or search for hidden information

Data mining:

→ we apply a ML algorithm on data and create a prediction model
ex: email - spam classifier

→ After applying ML to data and we are able to extract important data from the information called **data mining**.
DATE: / /

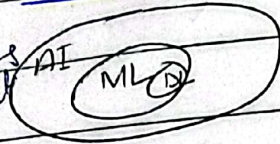
History of ML:

→ After 2010 it was stable. It remained significant data but in that era there was no data - in efficiency of hardware to run algorithms on such data.

- After evolution of internet 2010

AI VS ML VS DL VS Generative AI

more about parent content



- AI :- related to pattern recognition. Build apps that can perform its own task without human intervention.

ex: Netflix - Recommendation system like movies

Generative AI:

- subset of DL

- Two types of model training

use in data

Science.

models

streaming platform suggestion.

Self driving cars.

ML - provide set tools to

perform various task, statistical analysis, visualization, prediction

forecasting. To perform on data

DL - Mimic human brain.

Discriminative

→ Tasks:

Classification

Prediction

Trained on label dataset.

Generative multilayered NN.

AI → ANIV → Train machine

help to → CNN and object detection

create new content based on what it has already been used in LLM models.

→ RNN and its variants

Transformed & Boost

for text related series.

- Generates new data from trained data.

Generative AI

Generative AI have LLM, LIM, etc.

[LLM, LIM]

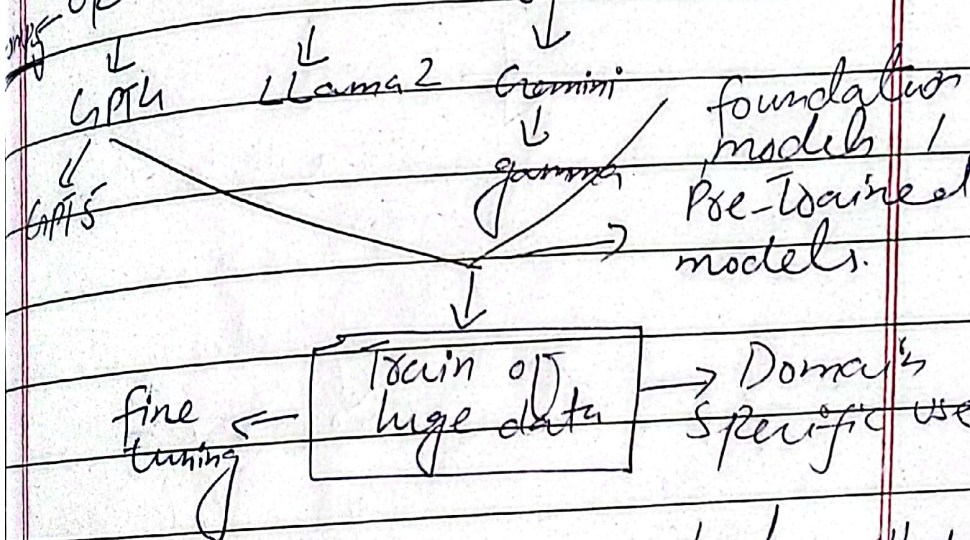
Text → image

What generative AI basically doing?

→ we have model that has already trained on huge amount of data and the tasks of these specific models are based on any if it generate a new data.

LLM models:

OpenAI, meta, Google, anthropic



Langchain: framework which will be able to work with the specific models

↓ RAG App.

ex of GAI: LLM

DL algorithm استعمال کے لئے ML سے
 DL سے متاثر ہوئے ہیں اور Biology سے inspired

Why DL?

DATE: 1/1/20

features of DL system
- saw just p. of DL

hidden layers of neurons

prediction of efficiency of system

data for models of ML
accuracy in point
is accuracy

data for DL
Prediction

Summary

ML ka matlab ye hai ki hum machine ko teach karte hain

like data/experiences

instead of telling computer exactly what to do step by step

we give it lot of data/examples and it learns to make its own prediction

- like training a smart robot that learns from examples instead of following rules

Algorithms → the computer uses to learn from data

helps the machine learn from example understand patterns
- make guess

Example: You want to teach your robot how to recognize fruits. You show it pictures of apples & bananas.

The algorithm is like teaching a trick. The robot uses to figure out differences. It looks at things like color, shape, and size. Then it uses what it learned to guess new fruit.

The o/p of ML is model. that represents the learned patterns and can be used for prediction & decisions.

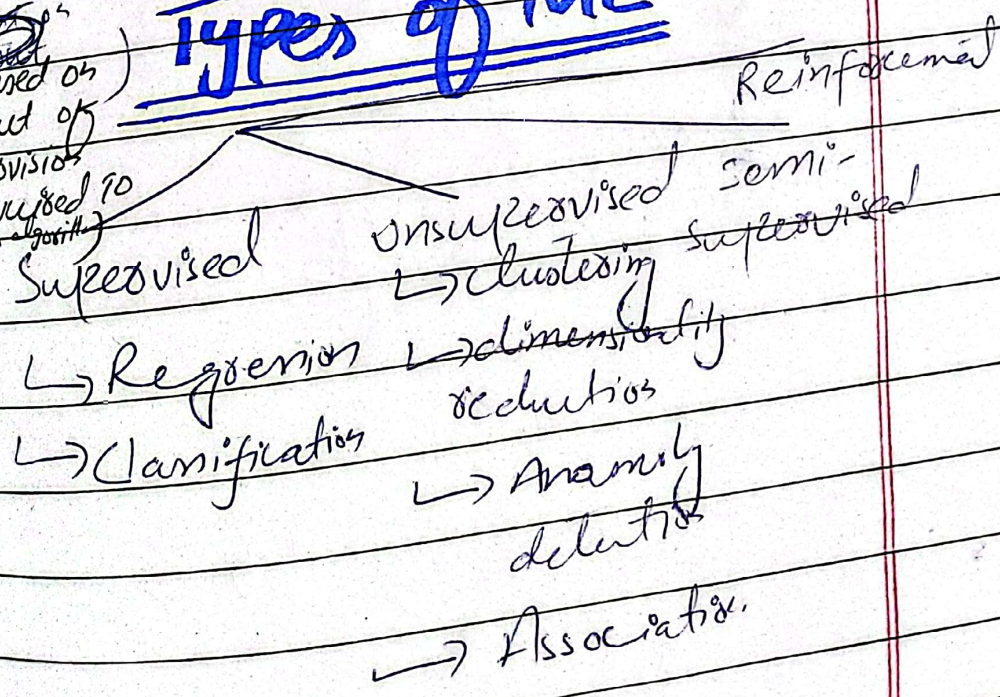
AI: refers to the development of program that behave intelligently and mimic human intelligence through a set of algorithm.

- focuses on these skills:

- 1) learning
- 2) reasoning
- 3) self-correction

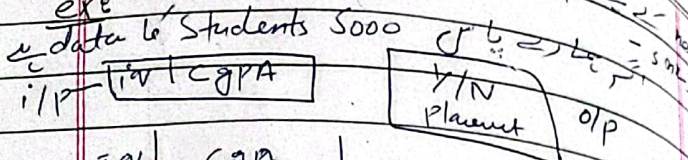
based on amount of supervision required to train algorithm

Types of ML



1. Supervised Learning:

DATE: 1/12/20
So that find relationship between i/p & o/p
ex: data of Students 5000



I/V	CGPA	Y/N	Placement
87	7.1	Y	New data
11	8.7	X	773, 734
75	6.3	N	Y/N

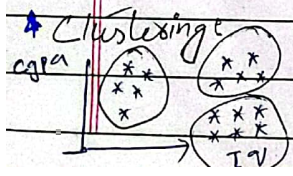
Regression: o/p & i/p are numerical
data → Numerical
→ Categorical
ex: I/V CGPA Package
8.7 8.9 6
11 8.8 4

Clustering: Categorical o/p & i/p
clustering → Categorical o/p & i/p
→ Problem

2. Unsupervised Learning:

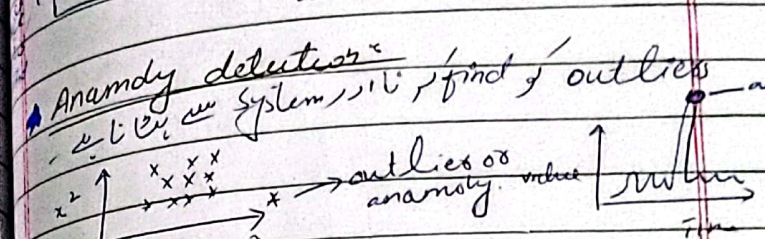
o/p & i/p are not given
ex: IQ | CGPA → Then it is unsupervised

we do in USL:
1) clustering 2) Dimensionality reduction 3) Anomaly detection 4) Association Rule Mining



Dimensionality reduction: slow algorithm
DATE: 1/12/20
Sometimes remove feature which is not useful
ex: room, washroom, sqft area
create new column by feature extraction
PCA algo will use

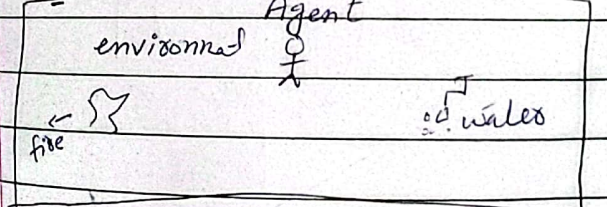
Anomaly detection: find outliers
DATE: 1/12/20
outliers or anomaly value
x-axis, y-axis, time



Association / Rule Learning: dependency of data item
DATE: 1/12/20
How to arrange items to make profit

3. Semi-Supervised: Partially supervised & unsupervised

4. Reinforcement Learning: learning by doing
- Not give any data by doing



Count: 0
 50 Pairs
 122
 000
 write

Handwritten notes and diagrams at the top right of the page.

1. observe
2. select action using policy

3. Action
4. (not reward or penalty)
5. update policy

Batch VS online ML

(based on how to train model)
 what is production? The server - on which your code is run.

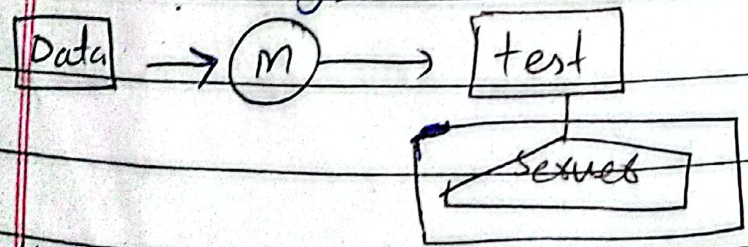
⇒ The environment of server where the code is run, called production env.
 ⇒ when code, called development environment.

⇒ ML behave different on different servers.

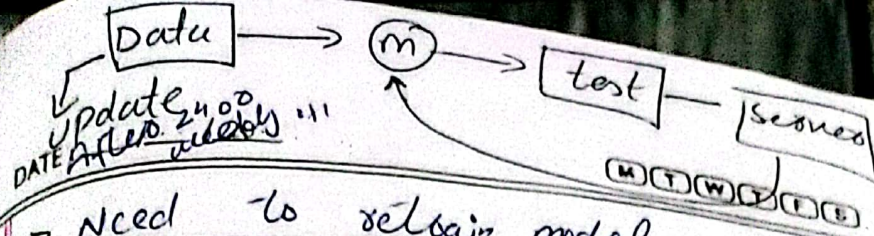
what is batch learning?

- conventional way of training ML model
- The whole data is used to train ML model.
- there is no incremental train
- mostly do offline on your machine.

Machine Learning Data
 Production



⇒ Here the model is static, once the model is trained new it will not be changed.
 - called batch learning problem.

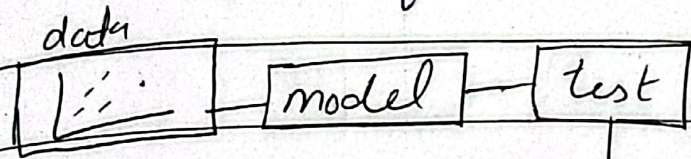


→ Need to retrain model again & again

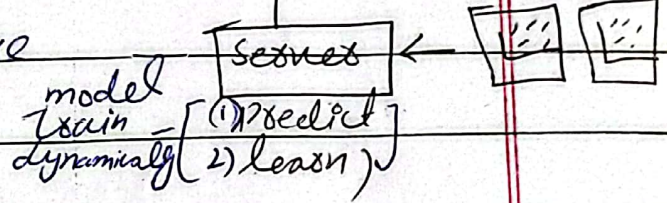
What is online ML?

مثلاً، Promotion کے Product پر ایسی مانی کی جائے گی کہ
 feasibility کم aurodoling - جی جی جی، Performance کے Product پر
 Performance improve. → online learning

- is done incrementally
- feed data in small batches to model
 these batches ^{form of} called mini-batches
 model improve after each batch.



ex: - chatbots are
 online ML.
 - swift key board
 - Youtube

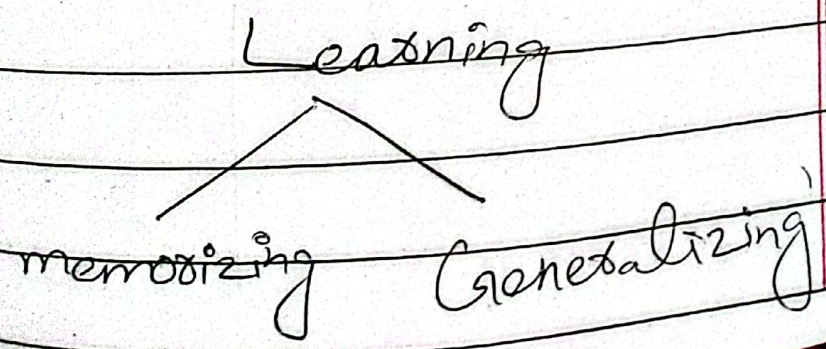


When to use both?

- if nature of problem is volatile
 then use online.

Types of ML based on how ML model learn

ML model learn by two @ ways: instance & model based



ایک اس طرح کے algorithm جو کہ
data کو instance سے
DATE کی help سے
dox line

MTWTFSS

DATE: 14/11/2020

دوسرے الگو کا ہم

Pattern so underline

entrant

Also called long reasoning: need to keep track of data

↑ Also called Instance based Placement Y/N?

Instance

ex	ID	CGPA	Placement Y/N?
	80	8	Y
	70	7	

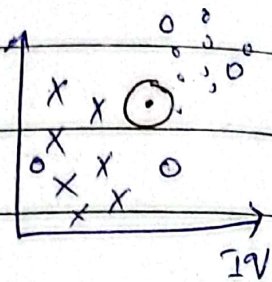
7.5 CGPA

X - No
O - Yes

103 CGPA

X X O

X - No
o - Yes



New: جو انٹرنیٹ کا باقی
 انٹرنیٹ سے distance (نفا) زیادہ
 اور جو فی سے میل جو انٹرنیٹ لین
 کے اگر جو انٹرنیٹ اینٹ سے
 New: جو انٹرنیٹ ایسی
 category سے belong کرے گا۔

ex³ KNN

Learning or Learning :- Instance based
- i.e. Concept

model based learning: just store ex
IQ / CGPA / Placements No need to train data

IV / Cgpa / Placed No need to give date

$$Y/N \quad \text{срн} \rightarrow Y$$


data points : Algorithm = Learning based
 things & variables behavior
 mathematical function

DATE: 1/120

MTWTFSS

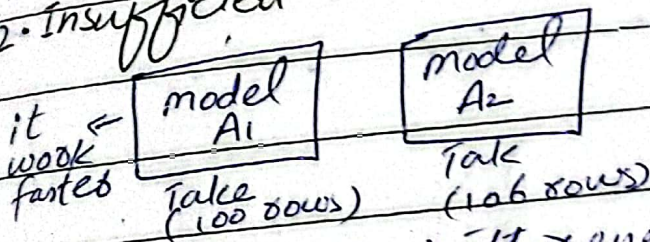
Mostly algorithm based on this
 find mathematical st's b/w it & o/p
 and draw decision boundary as
 mathematical function by using this
 decision incoming point can be classify.
 ex: Linear regression, Logistic,
 decision

Challenges in ML

1. Data collection: API
 ways: web scraping

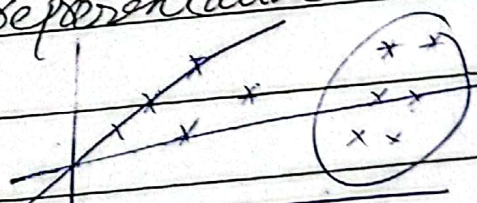
- but also problems

2. Insufficient data/labelled data



-> but A2 work better even though A1 is better.

3. Non-representative data: or sampling bias



4. Poor quality data.

6. irrelevant features:
 Garbage in, garbage out

7. Overfitting:

Train data & model ML on it
 < 100 features > 1000 data new data
 for prediction performance on new data
 is overfitting

7. Underfitting: opposite of overfitting
 DATE: 1/3/2022 Training data ~ M T W T F S S
 Unseen test ~ 1 - 100

Reasons:

- The model is too simple.
- Using too few features or wrong.

8. Software integration:

→ putting model into software.

each system has its own OS.

- bcz each ml model do not work well for each platform.

9. Offline Learning / Deployment

10. Cost involved

Applications of ML

1. Retail - Amazon / Big Bazaar
2. Banking & finance.
3. Transport - OLA
4. Manufacturing - Tesla
5. Consumer internet - Twitter

Machine Learning development Life Cycle?

- MLDC / MLDC

1. Frame the problem

- correctly frame the problem
- whether ml model is supervised or unsupervised? etc.

ETL

2. Gathering data

- Kaggle

- directly from csv file

- API - json form

- web scraping

- DB → DWH → Data Lake

- Spark clusters

3. Data preprocessing:
- what to do before processing
 - Remove:
 - 1. outliers
 - 2. missing data
 - 3. outliers
 - scale data.

4. Exploratory data Analysis:

- Try to study r/s between i/p & o/p.
- Entail r/s between data.

- Do visualization.
- Do univariate analysis
- Do bivariate analysis
- do multi-variate analysis
- outliers detection
- correct imbalance data to balance dataset.

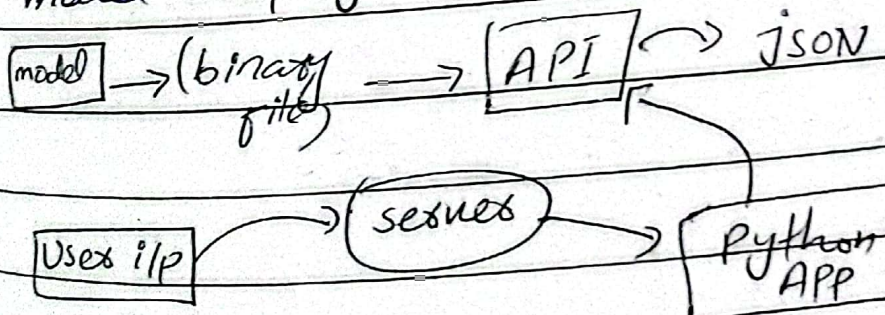
5. Feature engineering & selection

- ↓ i/p
- Sometimes create new columns for better analysis
- make changing in existing feature

6. model Training, Evaluation & selection

- do Ensemble learning
- ↓
- Combine multiple model parameters of model.
- and make 1 big powerful model.

7. model deployment



8. Testing

Beta/user Testing

9. Optimize

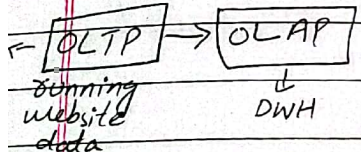
- Backup of model
- Backup of data
- Load balancing

how to Retrain? Rotting

Data Engineers vs Data Analysts

1. Data Engineers

- Job Roles:**
 - Scrap data from the given sources
 - Store the data in optimal servers / warehouse
 - Build data pipelines / APIs for easy access to the data
 - Handle db / DWH
- Skills Required:**
 - Strong grasp of algorithm & OS
 - Programming language (Java / Python & script writing)
 - Advanced DBMS
 - Big data tools (Apache Spark, Hadoop, Apache Kafka, Apache Hive)
 - Cloud platforms (Amazon web services, Google Cloud platform)
 - Distributed systems
 - Data pipeline



2. Data Analyst:

- Cleaning & organizing data**
 - Analyzing data to derive insights
 - Creating data visualization
 - Producing & maintaining reports
 - Collaborating with team
 - Optimizing data collection procedure

Skills: Statistical programming, Creative & Analytical thinking
 DATE: 1/12/20
 M T W T F S S

- Strong Communication skills
- Data mining - Cleaning & analyzing
- Data storytelling
- SQL
- Advanced MS excel

3. Data Scientist

- Full stack developer
- Do anything
- Make something for future

4. ML Engineer:

- Responsibilities:**
 - Deploying ML models to production
 - Scaling or optimizing model
 - Monitoring & maintenance of deployed models

- Skills:**
 - Mathematics
 - Programming language
 - ML model
 - Distributed system
 - ML model & evaluation
 - ML model

What are Tensors?

- are containers of data structure used in ML by ML libraries
- store numeric data
- most of 99.9% store numeric data

Tensor	Axis of Tensor
1. 0D Tensor / Scalars - array called 0D Tensor. eg 0.988(1)	same as 1D
2. 1D Tensor / Vectors [1, 2, 3, 4]	no of axes = rank = dim
3. ND array is called Tensor	

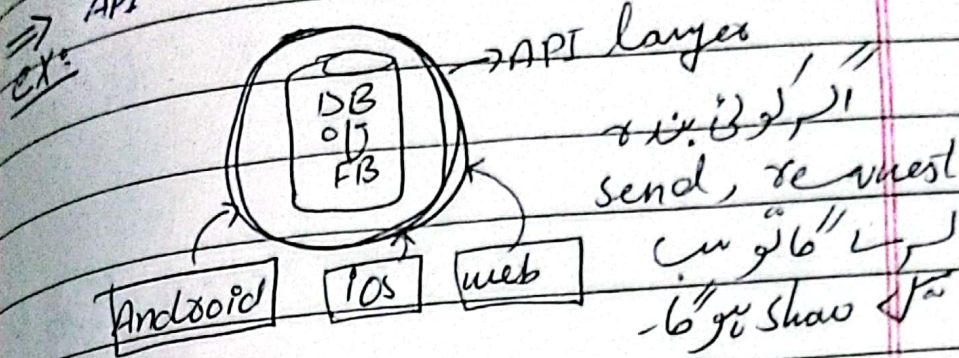
4: Fetching data from APIs

(M T W T F S)

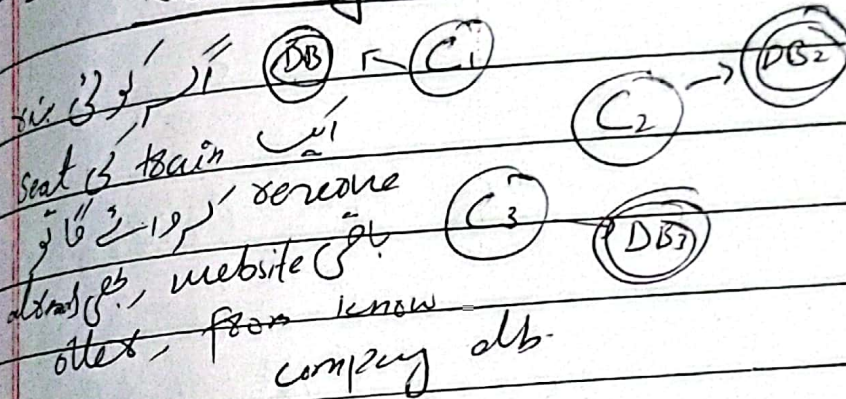
Q: What is API?

⇒ Enable communication between two software.

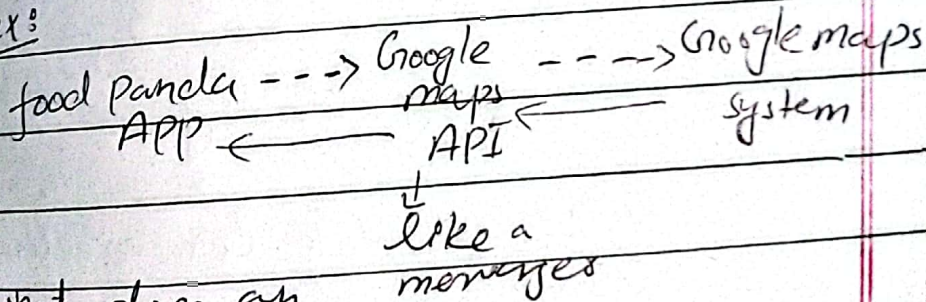
APIs are data pipelines.
API returns data in JSON form.



ex: Railway Reservation



ex:



Q: What does an API contain?

- What you can ask?

- How to ask?

- What format the answer will be?

Summary:
 - API is a messenger between your app and another service.
 - You send requests and get responses.
 DATE: 1/1/20
 API save time, not need to build everything from scratch.

REST API: Representational State Transfer. Works over internet over HTTP. ex of web API.
 => All REST API use API, but all APIs are not REST APIs.

Step 3: Understanding Your Data

- => Asking basic questions.
- Questions Ask?
1. How big the data is?
=> `df.shape`
 2. How does the data look like?
=> `df.head`
=> `df.sample(5)`
 3. What is the data type of cols?
=> `df.info`
 4. Are there any missing values?
=> `df.isnull().sum()`
 5. How does the data look mathematically?
=> `df.describe()`
 6. Are there duplicate values?
=> `df.duplicated().sum()`
 7. How is the correlation b/w columns?
=> `df.corr()['col-name']`

EDA using Univariate Analysis:

DATE: 1/1/20
 Analysis of single variable or single column.
 => Categorical data value column.
 => Countplot or Pie chart for visualize.
 Numerical data value column.
 => Histogram - show distribution of data
 a) Histogram - show distribution of data
 b) Displot - show probability of data
 c) Bonplot - provide five number summary
 => check for noise or outliers

EDA using bivariate and multivariate analysis:

- Analysis of two columns & more than two columns.
1. Scatter-plot (Numerical - Numerical)
 2. Box plot (Numerical - Categorical)
 3. Box plot (Numerical - Categorical)
 4. Displot (Numerical - Categorical)
 5. Heatmap (Categorical - Categorical)
 6. Clustermap (Categorical - Categorical)
 7. Pairplot
 8. Lineplot (Numerical - Numerical)

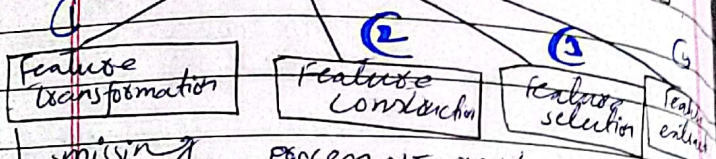
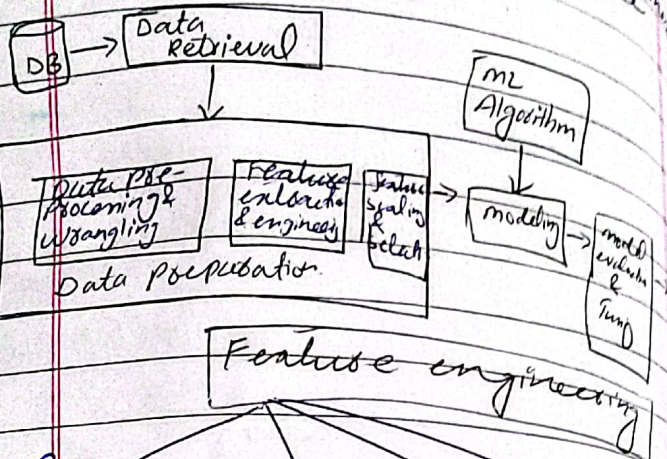
Pandas Profiling

Library for generating reports for EDA.

Feature Engineering

DATE: / /

Process of using domain knowledge to extract features from raw data. Features can be used to improve the performance of ML algorithms.



1. missing value imputation ⇒ process of replacing missing values with substituted values or either remove those values.

Handling categorical data

outliers detection

feature scaling

1. missing values imputation

→ mean/median/mode.

o numerical data → Use mean & median

o categorical data → Use mode

→ forward/backward fill

→ fill using previous or next valid observation

→ Constant value imputation

→ fill with a fixed number

→ KNN imputation

→ multivariate imputation

2. Handling categorical data

→ Since categorical data is not numerical, it needs to be converted into numerical form.

EX	index	Animal	on-hot side	index	log	cat
0	0	Dog	0	0	0	0
1	1	cat	1	1	1	1

3. outliers detection

→ values vary from normal ranges, called outliers.

4. Feature scaling

→ technique to make sure all the features are on the same scale or range.

2) Feature Construction

→ process of locating new features from the existing ones in dataset.

⇒ Goal is to make the data more useful for the model by adding new info that helps the model learn patterns better.

Purpose:

o improves model performance

o captures hidden patterns

o makes the model smarter

ex: $bmi = \frac{weight}{height^2}$

3) Feature Selection

selecting a subset of relevant features to simplify models & improve performance.

Purpose:

o Reducing dimensionality

o Improving model accuracy

o Reducing overfitting

o Improving model interpretability

Techniques: o filter methods o wrapper methods o embedded methods

1. Feature Extraction

⇒ from given columns, create new features.

⇒ we create new features by using selective features from new data.

Feature Scaling

⇒ Technique to standardize features present in the data in a way that they will work well. If scale is not same then there will be a problem.

ex: KNN not work well.

Types of FS

Standardization
Also known as z-score Normalization.

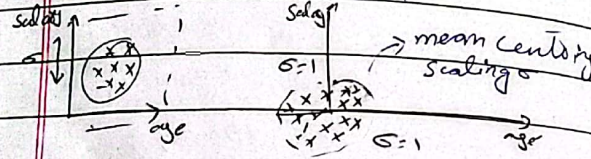
Normalization
- min-max scaling
- mean normalization
- min absolute
- Robust Scaling

1. Standardization:

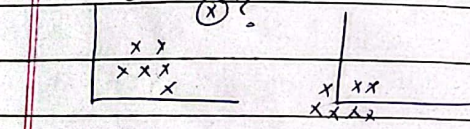
$$\text{formula: } x_i = \frac{x_i - \bar{x}}{\sigma}$$

ex: age → age' → mean of new number will be zero

$\mu = 0, \sigma = 1$



Impact of outliers? When to use SD?



Note: Before feature scaling better first train-test-split.

Class in sklearn for doing standardization from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaler.fit(x_train) # calculate μ & σ

transform

$x_{train_scaled} = \text{scaler.transform}(x_{train})$

$x_{test_scaled} = \text{scaler.transform}(x_{test})$

Note: fit → always training data

about transform both train & test data.

Note: StandardScaler takes df but returns numpy array.

⇒ need to convert again in df

$x_{train_scaled} = \text{pd.DataFrame}(x_{train_scaled}, \text{columns}=x_{train}.\text{columns})$

⇒ same like for x-test scaled.

⇒ For checking scaling happened do:

$\text{np.round}(x_{train_scaled}.describe(), 1)$

$\text{np.round}(x_{train_scaled}.columns)$

⇒ Note: After scaling or standardization shape of distribution will never be change only scale, μ change.

Why use? Logistic regression algo work better.

Note: StandardScaler 16-20 outliers (outlier for 1% if impact ≤ 0.1 for 9% same outliers if we need to handle explicitly - 20%.

When to use?

When use the following algo

→ K-means, KNN, PCA, ANN, Gradient Descent

2. Normalization:

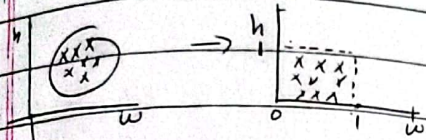
- Applied as part of data preparation to make the numeric columns just keep from losing info that is on the common scale, without introducing bias.

1. Min-Max Scaling

$$\text{form: } X_i' = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}}$$

$\Rightarrow$ Apply on each value of column.

$\Rightarrow$ Note: the range of new distribution will be $[0, 1]$.



$\Rightarrow$ Use MinMaxScaler in Scikit Learn to do this.

2. Mean Normalization:

$$\text{form: } X_i' = \frac{X_i - X_{\text{mean}}}{X_{\max} - X_{\min}}$$

range of o/p values = $[-1 \text{ to } +1]$

$\Rightarrow$ useful in algorithms where categories need, but rarely use and not support by sklearn.

3. MaxAbsScaling:

$$\text{form: } X_i' = \frac{X_i}{|X_{\max}|}$$

$\Rightarrow$ Claim in Scikit Learn: MaxAbsScaler

$\Rightarrow$ useful where data is sparse.

Robust Scaling

$$\text{form: } X_i' = \frac{X_i - X_{\text{median}}}{\text{IQR}}$$

$\Rightarrow$ Claim available in Scikit Learn: RobustScaler
 $\Rightarrow$ useful when data has lot of outliers.

$$\text{IQR} = Q_3 - Q_1$$

$$Q_1 = (n+1)^{\text{th}} \text{ term} \times 1/4$$

$$Q_3 = \frac{3}{4} \times (n+1)^{\text{th}} \text{ term}$$

Normalization VS Standardization

1. Is feature scaling or mixed?
2. Better result mostly meet in standardization.
3. min max useful when min-max values also known.
4. when no idea use standard scaling.
5. normalization change the units of data - units are lost, result is unitless.
6. standardization also change unit. unit becomes the standard deviation.

Encoding Categorical data

Categorical data

Nominal
 $\Rightarrow$ Not have any order
ex: state

- categories that cannot be ranked.
- ex: colors, genders, Types of animal.
- There is no meaningful...

ordinal

$\Rightarrow$ order between categories
ex: Taking position

- categories have meaningful order and scaling.
- ex: education level.
- Customer satisfaction, rating scale.

⇒ Since ml algorithms require numbers
So, encoding is required.

DATE: 1/12/20
1. Ordinal encoding

X	Y

Note: ordinal encoding is used for data with a natural order. Use label encoding for data without a natural order.

ex:

education: HS, PG, UG, UG, HS

{ PG → 2, UG → 1, HS → 0 } tell explicitly

2. One-Hot encoding

- Use for handling nominal categories

data:

X	Target
Colo	0
Yellow	1
Blue	1
Red	0
Yellow	0

Colo-Y	Colo-B	Colo-R
1	0	0
0	1	0
0	0	1
1	0	0

[Y, B, R]
0, 1, 2

Note: If categories are not ordered, use one-hot encoding. If ordered, use ordinal encoding.

⇒ n-column but we remove 1 col

⇒ n-1 columns

⇒ 1st col just remove from dummy cols

⇒ multicollinearity

ColumnTransformer:

⇒ Scikit learn has this class.

⇒ Sometimes we need to perform separate encoders on each column, and then combine all columns that will consume lot of time. To overcome this we use ColumnTransformer class.

transformer = ColumnTransformer([transformer_list])
= ['transformer_name', transformer_name, [col, col, ..., n], ...]
remainder = 'passthrough or drop'

ex: cal	num	cal
1	1	NA
2	2	NA
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8	8	NA
A	NA	A
B	NA	B
C	NA	C

Handling missing data

Done already.

Gradient Descent

- ⇒ Optimization technique.
- ⇒ Method to find the best model by reducing the error step by step.
- ⇒ Like going downhill to reach the lowest point where the prediction error is lowest.

$$\text{form } \theta = \theta - \eta \frac{dJ}{d\theta}$$

θ → param like weight slope
 η → learning rate or step size
 J → cost function

Types:

1. Batch GD - use all training data at once
2. Stochastic GD - use one random training data.
3. minibatch - use small batches of training data.

ML Algorithm:

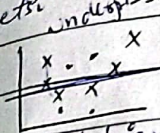
1. Linear Regression
2. Multiple Regression
3. Polynomial Regression

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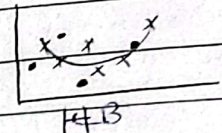
Bias variance Trade off
 ⇒ Concept in ML that explain how to balance underfitting & overfitting

- Bias: Inability of ML model to truly capture the θ s in the training data.
 High bias = inaccurate model.

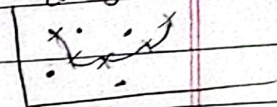
- Variance: How much a model's predictions change when trained on different data sets.
 underfit - simple
 overfit - complex



High bias
 LV



LB
 LV



Low bias - Test data
 High var - Training data

overfitting: model best on training data
 - training error ↓ but testing error ↑
 So, variance is very high - High variance

Underfitting:

- Not best on both data.
- High bias

Target / Trade off

- Low bias • Low variance

Regularization adding some term to loss function.

L2: Ridge reg: $MSE + \lambda ||w||^2$

L1: Lasso reg: $MSE + \lambda ||w||_1$

also known as least absolute shrinkage & selection operator. Also used to select imp features.

Code: from sklearn.linear_model import

note: as the value of λ is increased, the slope will be down.

Lasso regression can set some coefficients zero, effectively selecting the most imp features.

- Use cases of Lasso: for high dimensional data.

- for obtain a sparse model with only a few non-zero coefficients.

why Lasso creates sparsity?

& why Ridge not creates?

for Lasso: $b = \bar{y} - m\bar{x}$

$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda |m|$

$\frac{dL}{dm} = \sum_{i=1}^n (y_i - m\bar{x} - \bar{y} + m\bar{x}) + \lambda$

$= 2 \sum_{i=1}^n (y_i - m\bar{x} - \bar{y} + m\bar{x}) + 2\lambda$

$(-2 \sum_{i=1}^n (\bar{x} - \bar{y})) + 2\lambda = 0$

$= -2 \sum_{i=1}^n ((y_i - \bar{y}) - m(\bar{x} - \bar{x}))(\bar{x} - \bar{x}) + 2\lambda = 0$

$= -2 \sum_{i=1}^n ((y_i - \bar{y})(\bar{x} - \bar{x}) - m(\bar{x} - \bar{x})^2) + 2\lambda = 0$

$= -2 \sum_{i=1}^n ((y_i - \bar{y})(\bar{x} - \bar{x}) - m(\bar{x} - \bar{x})^2) + 2\lambda = 0$

Elastic Net Regression: Combination of

DATE: / /